University of Rochester Medicine

Table of Contents

Introduction to Statistical Computing, BST 430, ... 3

Fall 2026 NYSED **..... 3**

Instructor Information 3

TANZY M. LOVE, PHD NYSED **..... 3**

Donald Harrington, M.S. 3

E-mail: Donald_Harrington@urmc.rochester.edu 3

Hongyue Cookie Wang, Ph.D. 3

E-mail: Hongyue_Wang@urmc.rochester.edu 3

Shan Gao, M.S. 3

E-mail: Shan_Gao@urmc.rochester.edu 3

Contact Information and Preferences 3

OFFICE HOURS 3

Teaching Assistant Names and Responsibilities 4

Teaching Assistants Contact Information and Preferences 4

Course Details 4

MONDAYS AND WEDNESDAYS, 11:00-12:40 4

SAUNDERS RESEARCH BUILDING 1.404 NYSED 4

Course Format and Modality 5

COURSE WEBSITE/LINK TO BLACKBOARD 5

Course Description NYSED 5

Basic/intermediate R programming; statistical analysis in R; visualization in R; reproducible research and collaborative coding; command line tools and BlueHive; introduction to SAS programming; statistical analysis in SAS. Topics in statistical analysis provide working examples. Prerequisite Courses NYSED 5

Credit Hours NYSED 5

Bibliographic or Other Resources NYSED 5

Course Objectives and Learning Outcomes 6

Purpose Statement 6

Course Objectives (2 – 3 objectives) NYSED 6

Course learning Outcomes (3 – 5 outcomes) NYSED 6

Learning Activities and Assignments 7
LEARNING OVERVIEW – MAJOR TOPICS 7
Graded Assignments 8
Grades and Related Policies 8
Success Statement 8
Methods for Assessing Student Achievement 9
Feedback Statement 9
Final Exam Date 9
Late and Make-up Work Policy 9
Basis of Grade Determination 10
Academic Honesty Policy 10
Generative AI Policy 11
Other Policies and Expectations *11*
Communication 11
CLASSROOM CULTURE AND ENVIRONMENT 11
Accessibility and Academic Accommodations 12
https://www.rochester.edu/disability/ 13
Religious Holidays 13
Academic Calendar and Registrar Policies 13
Anti-discrimination and Harassment 13
Title IX 13
Student Resources 14
Course Schedule *15*

Introduction to Statistical Computing, BST 430, Fall 2026

Instructor Information

Tanzy M. Love, PhD

Please call me Tanzy (tan-zee) or Dr. Love.

Donald Harrington, M.S.

E-mail: Donald_Harrington@urmc.rochester.edu

Hongyue Wang, Ph.D.

E-mail: Hongyue_Wang@urmc.rochester.edu

Shan Gao, M.S.

E-mail: Shan_Gao@urmc.rochester.edu

Contact Information and Preferences

tanzy_love@urmc.rochester.edu

The best way to communicate with me is by email. It can take me 1-2 business days to respond to email. Please note that I do not check email regularly after 5:00 pm or on the weekend.

Dr. Love is the instructor of record, but please reach out to other faculty directly for help with the material that they cover.

Office Hours

Office hours only by appointment. Please email me at least 24 hours before you want to meet to discuss homework or other course questions. I will generally be available at the end of class to stay and answer questions.

If you have questions related to your grade in the course or of a personal nature, please email me for a private appointment rather than bringing these sensitive topics to a public place where other students may be present.

EMERGENCY AND SENSITIVE COMMUNICATION

If you need to communicate something of a personal or sensitive nature, please contact me directly via email at tanzy_love@urmc.rochester.edu *to set up an appointment. I will be happy to meet with you.*

TEACHING ASSISTANT NAMES AND RESPONSIBILITIES

James Hopham - grading

Lucy Langenberg - grading

TEACHING ASSISTANTS CONTACT INFORMATION AND PREFERENCES

james_hopham@urmc.rochester.edu and *lucy_langenberg@urmc.rochester.edu*

Please communicate with the TAs by email. It can take 1-2 business days to respond to an email. Please note that TAs do not check email regularly after 5:00 pm or on the weekend.

Office Hours: Held in SRB 4107

James: Mondays 12:40-1:30 pm

Lucy: Wednesdays 2:00-3:00 pm

COURSE DETAILS

MONDAYS AND WEDNESDAYS, 11:00-12:40

SAUNDERS RESEARCH BUILDING 1.404

Monday	Tuesday	Wednesday	Thursday	Friday
11:00 – 12:40 am SRB 1.404 Class		11:00 – 12:40 am SRB 1.404 Class		
Complete: Assigned reading Learning journal		Complete: Assigned reading Learning journal		

Course Format and Modality

Instruction will be primarily face-to-face.

Course website/link to Blackboard

https://rochester.box.com/s/1w5b8i02elogauofyjofiexuljrnj8by (SAS session)

https://bst-urmc.github.io/bst430-fall2026-site/ (not live yet)

https://learn.rochester.edu (up and running)

Course Description NYSED

Basic/intermediate R programming; statistical analysis in R; visualization in R; reproducible research and collaborative coding; command line tools and BlueHive; introduction to SAS programming; statistical analysis in SAS. Topics in statistical analysis provide working examples.

Prerequisite Courses NYSED

There are no prerequisites for this course.

Credit Hours NYSED

Credits allocated: 4 credits. This course will include 150 minutes per week of lecture time (two lectures), 50 minutes per week of lab time (one time per week), and 400 minutes per week of out-of-class student work (e.g., reading assignments, writing, research, problem sets, study time, and other unsupervised work).

Bibliographic or Other Resources NYSED

Part I:
Textbook: The Little SAS Book, 5th edition by Delwiche and Slaughter
Part II:
Wickham & Grolemund "R for Data Science" https://r4ds.had.co.nz/

Recommended texts, Part II
Peng "R Programming for Data Science" https://leanpub.com/rprogramming
Wickham "Advanced R" https://adv-r.hadley.nz/
McBain "Git for Scientists" https://milesmcbain.github.io/git_4_sci/
Wickham "R Packages" https://r-pkgs.org/

Required software and programming languages

Part I:

Required Software: SAS version 9.2 or higher or SAS Studio

Course Materials and assignments will be stored in a Box folder

Part II:

We will be using RStudio in class examples; it is free. You must have a working install of R 4.6+ (free). If you don't want to work on your own computer, you can use the cloud version provided by the DBCB, though the instructor is becoming more familiar with it there are staff who can help you get it set up. You should bring a laptop to class every day, as we will regularly hold in-class labs. These labs will be collaborative, so hopefully most students will have laptops.

COURSE OBJECTIVES AND LEARNING OUTCOMES

PURPOSE STATEMENT

This course aims to prepare students to use computers to implement a variety of statistical methods. At the end of this course, each student should be able to work as a consultant, grade as a TA, and organize code for their thesis.

COURSE OBJECTIVES NYSED

In this course, we learn the most popular statistical software (1) SAS and 2) R) for data management, analysis, and visualization. We also learn the basics of computer coding for statistical collaboration and research.

COURSE LEARNING OUTCOMES (3 – 5 OUTCOMES) NYSED

By the end of this course, you will be able to:

1. ***Read** data in/out of SAS and R databases*
2. *Sort, subset, merge, and **reformat** databases*
3. ***Create** useful figures*
4. ***Collaborate** on code in GitHub*
5. ***Make** reproducible code with comments/annotations and readable style*

LEARNING ACTIVITIES AND ASSIGNMENTS

LEARNING OVERVIEW – MAJOR TOPICS

Part I:
• Create and run SAS programs in a PC environment or SAS Studio
• Read raw data files in various formats and create SAS data sets
• Create new variables in the data step
• Use SAS procedures to describe data numerically and graphically
• Annotate SAS output with informative titles, labels, and formats
• Work with SAS data sets: sort, subset, merge, and re-format
• Use commonly used SAS procedures for statistical inference and modelling
• Export SAS data and output to other computers and software
Part II:
• Be able to utilize and recognize common programming concepts and constructs
• Collaborate and share code using git and github
• Create useful figures
• Implement reproducible data analyses in Rmarkdown
• Write documented and maintainable code for common statistical tasks
• Effectively debug and test code for correctness

LEARNING ACTIVITIES

You will engage with the course material through a series of activities, in and out of class. These activities are designed to provide you with opportunities to practice and receive feedback on the key skills and application of knowledge necessary to successfully achieve the course learning outcomes. You will be asked to:

- ***Read** 'reference' material and notes. You will be gathering information and beginning to build a mental toolkit. For the rest of your life, most of your learning will be by reading instead of by lecture.*
- ***Homework.** You will have problem sets to practice the analytical and coding techniques taught in the class.*
- ***Labs.** You will have an opportunity to work together to try the techniques that you have learned in class on more complex problems with help from the professor.*

GRADED ASSIGNMENTS

Homework – this assignment will be given approximately every two weeks and due two weeks after it is assigned. The purpose of the problem sets is to practice the coding techniques we have discussed in class and shown in the notes. The questions on the problem sets are examples of performing statistical computing. In general, these questions have a correct answer, though there are often multiple ways to derive the correct answer. ***Some questions will ask for interpretation or description.*** *Do not just turn in code for these. These questions will be graded on how well you support your argument and whether the numbers and plots that you have provided agree with your conclusions. Coding will also be graded on style (after style is covered) and on GitHub commit statements and comments in the code.*

Labs– this assignment will be given approximately every two weeks and due two weeks after it is assigned (the first two are short and you will only get a week for each one). The purpose of the labs is to practice working together and creating a finished product. ***Only one lab report needs to be submitted per group.***

What:	When:	How often:	Instructions:	How to submit:
SAS Homework	TBD	every week	Box folder	Email
R Homework	5pm Mondays	every 2 weeks	Class GitHub webpage	GitHub .md
Lab reports	5pm Wednesdays	every 2 weeks	Class GitHub webpage	GitHub .md
Final exam	24 hours of final exam week	Once	Class GitHub webpage	GitHub .md

GRADES AND RELATED POLICIES

SUCCESS STATEMENT

This course, like many technical courses, is challenging and requires you to learn a lot of material and develop and apply complex knowledge and skills. There is rarely a simple or single correct way to answer the questions we are exploring and that can be frustrating. **I know that each and every one of you can be successful in this course** *and, as your instructor, I want you to have the tools you need to succeed. Therefore, if I notice you are struggling or if I haven't seen or heard from you for a period of time, I may message you with suggestions for ways to connect to appropriate resources. I ask that you respond to this outreach, whether or not you plan to use the support – responding to offers of support is an important skill to build.*

Methods for Assessing Student Achievement NYSED

Participation	Class meetings	5 pts	5%
SAS Assignments	3 assignments	25 pts	25%
R Assignments	4-5 assignments	25 pts	25%
Lab Reports	6 assignments	25 pts	25%
Final Exam	1 exam	20 pts	20%
Total		*100 pts*	*100%*

Feedback Statement

My goal is to provide you with feedback on your drafts and submitted assignments within 2 weeks of the due date. The TAs provide feedback and I supervise them. If we need more than the anticipated 2 weeks, I will make an announcement in class and send a message via Blackboard. TAs will provide you with feedback in GitHub on the document you submit, as comments and annotations.

Final Exam Date

There is a 24 hour take-home final exam in this class.

Late and Make-up Work Policy

You each have 2 late days that may be used on the following assignments without any consequence to the assignment grade: assignments. These late days require no explanation and should be used when you need extra time – for whatever reason – but do not need to be used in the case of illness or other events outside of your control. If you have a question about whether a situation qualifies as an excused lateness, please ask. ***To use late days****: notify me at the start of class the day the assignment is due. Late days start at midnight of the due date. If more than 2 late days are accumulated the assignment will be graded as late.*

Late assignments receive an automatic 25%-point reduction. It is always better to submit work late than not submit the assignment at all.

Please remember that New York Education Law provides that if you choose not to attend classes or take exams on certain days because of your religious beliefs, you will be given an equivalent opportunity to make up the work or exam without penalty. These occurrences are considered excused absences, and you should not use one of your 2 late day allowances in these instances.

Instead, please simply notify me by email in advance of the due date that your work will be late or that you will miss a scheduled exam and we will arrive at a new deadline/schedule together.

BASIS OF GRADE DETERMINATION NYSED

Grades for graduate students are reported on one of two systems per SMD's Grading Policy. A grade of E is considered a failing grade. Students do not receive academic credit for a failing grade and must retake any courses required for their program in which they receive a failing grade. Failing grades will be assigned to students who have not mastered course content sufficiently to progress.

Grades of B- or better are required for satisfactory performance in the PhD program. One grade below B- in any course or research experience is cause for review of overall performance and will result in academic probation for the duration of graduate study. A second grade below B- is basis for dismissal from the PhD program.

- A (excellent), A-, B+, B (good), B–, C (poor), and E (failure)
- S (satisfactory) and E (failure)

90-100%	*A (excellent)*
85-90%	*A-*
80-85%	*B+*
75-80%	*B (good)*
70-75%	*B-*
60-70%	*C (poor)*
<60%	*E*

ACADEMIC HONESTY POLICY NYSED

Academic honesty is important to the course, as it ensures a level playing field and safeguards the value of your grade. You can find information about the academic honesty policy in the University Graduate Student Regulations and Policies. Please follow these course policies and those for the use of AI listed in the next syllabus section.

1. *Students are encouraged to study together to complete the assignments and to study for the test, the writing and code in assignments should be individually-written.*
2. *Lab assignments are collaborative, there will be one grade for all members of the lab.*

3. *The final exam will be done without help from any other students and you need to be prepared to answer questions about it.*

I will specify any exceptions to these policies in writing. I'm always happy to answer any questions you might have about academic honesty; just ask!

Generative AI Policy

COURSE	EXAM	DISCLOSURE POLICY
Unless I Say No	***Thoughtfully***	***Citation + Description + Reflection***
Students are encouraged to use generative AI when helpful unless specifically forbidden for particular student work.	***All work related to completing the exam must be your own and you are responsible. Generative AI may be used, thoughtfully.***	Include a citation, description, and a reflection each time AI has been used in your work.

Other Policies and Expectations

Communication

See page one for information about how to communicate with me and the TA(s).

Class Announcements

Class announcements will be made at the beginning of each class and sent thru class list email via Blackboard. Please check your University email and Blackboard regularly for updates.

Classroom Culture and Environment

Learning Environment

Each voice in our classroom is important and brings with it a wealth of knowledge. Please honor the uniqueness of your fellow classmates and appreciate the opportunity we have to learn from each other. Please respect your fellow students' opinions and refrain from personal attacks or demeaning comments of any kind. If something is said in class that negatively impacts your learning (including by me!), please do not hesitate to let me know.

Community Norms and Expectations

As a community for and of learning, it is important that we have shared expectations for our time together. Please find below some general expectations for our time together.

You can expect me to:

- *Be passionate about your learning and success*
- *Cultivate a culture and classroom climate of mutual respect, active listening, critical thinking, and honest collaboration*
- *Provide you with thoughtful, constructive, and timely feedback on your assignments*
- *Show up prepared and on time for class meetings and office appointments*
- *Respond swiftly and effectively to your concerns, feedback, and/or suggestions*

I expect you to:

- *Actively participate and engage in the learning process through self-reflection and an open mind*
- *Contribute actively to the course culture of mutual respect, active listening, critical thinking, and honest collaboration*
- *Embrace the opportunity to learn from and with each other*
- *Submit thoughtful responses to assignments*
- *Communicate with me frequently so I know best how to support your learning and success in this course*

Attendance Policy

Regular attendance is essential for success in this course. More than three unexcused absences will result in loss of all your participation points. Excused absences include UHS-documented appointments, hospitalization, family emergencies or deaths, and/or serious personal concerns which impact your ability to attend class and/or complete assignments in a timely manner. Excused absences need to be reported to me via email at tanzy_love@urmc.rochester.edu as soon as possible and include documentation, if applicable.

ACCESSIBILITY AND ACADEMIC ACCOMMODATIONS

The University of Rochester welcomes applications from students with disabilities. The University employs professional staff committed to assisting students with disabilities in the classroom, residence halls, libraries, and elsewhere on campus. It is a personal decision to disclose the existence of a disability and to request an accommodation. A decision not to disclose will be

respected. Students seeking accommodation for a disability or temporary medical condition must request accommodation through the UR Office of Disability Resources.

https://www.rochester.edu/disability/

Religious Holidays

SMD policy and practice are that classes not be dismissed on religious holidays. It is expected, however, that students will not be penalized in any way for observing religious holidays. Instructors are asked not to schedule examinations on the same day as holidays. New York Education Law provides that students who choose not to attend classes or take exams on certain days because of their religious beliefs will be given an equivalent opportunity to make up the work requirements or exams they miss without penalties.
For a list of the major religious holidays, see the Interfaith Chapel's holidays page.

Sample language: I have done my best to ensure major course assignment due dates and exams do not fall on or during religious holidays. If an assignment or exam is scheduled to conflict with your ability to observe a religious holiday in your tradition, please let me know as soon as possible so that we can arrange an alternative date or assignment.

Academic Calendar and Registrar Policies

The SMD Academic Calendar provides important deadlines for drop/add/withdraw from courses, as well as schedules for final exams, the Rochester Monday that occurs once during spring semester, breaks and holidays, and University Commencement. The University Registrar's website also covers policies for audits, leaves of absences and withdrawal from the University, incompletes, and privacy.

Anti-discrimination and Harassment

Please see the Policy Against Discrimination, Harassment, and Discriminatory Employment/Service Practices website for the University anti-discrimination and harassment policy. You can report an issue on the Office of University Engagement and Enrichment website.

Title IX

Please see the Sexual Misconduct and Title IX website for the University Title IX policy. You can report an issue on the Office of University Engagement and Enrichment website.

STUDENT RESOURCES

The University of Rochester offers a wide range of academic, health, and student life resources to support your success and well-being throughout the semester.

See *https://www.urmc.rochester.edu/education/graduate/trainee-handbook/trainee-support-resources*

COURSE SCHEDULE

This course schedule is tentative and may change during the semester in response to extenuating circumstances, such as weather or illness, and/or to ensure successful learning and achievement of course learning outcomes and goals.

Week	Date	Lecture	Planned Material	HW/LB
1	31-Aug	S1	SAS environment, rules, variables, data step, input statements (Wang)	HWS1
1	2-Sep	S2	Data exploration with SAS (Gao)	
2	7-Sep		LABOR DAY	
2	9-Sep	S3	Data cleaning with SAS (Gao)	HWS2
3	14-Sep	S4	Data visualization with SAS (Gao)	
3	16-Sep	S5	Control output using ODS, Macros and debugging (Gao)	
4	21-Sep	S6	Statistical Testing/Modeling 1: TTEST, ANOVA, GLM, NPAR1WAY (Wang)	HWS3
4	23-Sep	S7	Statistical Testing/Modeling 2: PROC LOGISTIC, PROC GENMOD, PROC MIXED (Wang)	
5	28-Sep	1	Syllabus, Projects/repositories, rmarkdown, Intro to R (syntax)	HW1
5	30-Sep	2	ggplot (i)	LB1
6	5-Oct	3	dplyr (i) changing data	
6	7-Oct	4	Data types/structures in R and Collaboration with git/github	LB2
7	12-Oct	12	Functions (i) scope, if else, error messages	HW2
7	14-Oct	7	Factors and dates	LB3

8	19-Oct	9	Data from files (Harrington)	
8	21-Oct		AI for graduate level statisticians (Weaver)	LB3
9	26-Oct	14	Debugging (Harrington)	HW3
9	28-Oct	5a	Merging: Data frames in R	LB4
10	2-Nov	5b	dplyr (ii) advanced changing data	
10	4-Nov	6	Styleguide	LB4
11	9-Nov	10a	Text manipulation, Regular expressions (regex)	HW4
11	11-Nov	10b	Text analysis	LB5
12	16-Nov	11	Indexing, iteration, linear algebra	
12	18-Nov	13	Function (ii) classes, functional iteration, and generics	LB5
13	23-Nov	8	No class, watch the video Visualization: ggplot (ii), other graphing systems	
13	25-Nov		THANKSGIVING	
14	30-Nov	15	Linear models	HW5
14	2-Dec	16	Multivariate linear models and prediction	LB6
15	7-Dec	17	Logistic regression and prediction	
15	9-Dec	20	Machine learning: bagging and boosting	LB6
16	14-Dec		Special topic	
Finals			24 hour take-home Final Exam	FINAL

Last update: 2026 Oct 12